PLANT PEST

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Karnal bunt

Exotic to Australia

Features: A fungal disease that causes broken hollow (bunted) grain, powdery masses of dark spores, and a strong fishy odour

Where it's from: India originally, but now also Afghanistan, Brazil, India, Iran, Iraq, Mexico, Nepal, Pakistan, South Africa and United States

How it spreads: Spores spread easily on infected wheat seeds, soil, agricultural products, clothing, machinery and on the wind

At risk: Wheat, durum wheat, triticale

Australian Chief Plant Protection Officer, Dr Gabrielle Vivian-Smith, provides an overview of the risks posed by Karnal bunt.

Karnal bunt is a fungal disease caused by the fungus Tilletia indica. it infects grains at flowering.

It causes broken hollow or bunted grain with powdery masses of dark spores and a strong fishy odour.

The name Karnal comes from the city in India where the disease was first identified in 1931.

Since then, it has spread to Afghanistan, Nepal, Pakistan, Iran, Iraq, South Africa, Brazil, Mexico, and the United States.

Its spores spread easily on infected wheat seeds, soil, agricultural products, clothing, machinery and on the wind.

Contamination levels of as little as 3% in wheat is enough to render it unfit for human consumption.

If Karnal bunt made it to Australia, it would have a major trade and economic impact—over 45 international markets could reject our grains and grain prices would be negatively affected.

The fungus would be almost impossible to eradicate once here since its spores can persist in soil for up to four years and they can travel on the wind over long distances.

Import shipments may need to be treated and certified so make sure that you always check Biosecurity Import Conditions before you import anything into Australia.

Keep it out

Karnal bunt (also called partial bunt) is caused by the fungus Tilletia indica which infects grains at flowering. It reduces grain quality through the production of masses of powdery spores that discolour the grain and grain products. It is recognised by a fishy smell which taints the grain.

The name comes from the city in India, Karnal, where the disease was first identified. Unlike other bunt diseases, only some grains are affected on each wheat ear.

If Karnal bunt made it to Australia, it would have a major economic impact—over 45 international markets would reject our grains and grain prices would plummet.

The fungus would be almost impossible to eradicate once here since its spores can persist in soil for up to four years and they can be carried over long distances by wind.

Importing goods

To keep Karnal bunt out of Australia, never ignore Australia's strict biosecurity rules.

Import shipments may need to be treated and certified, so before you import, check our Biosecurity Import Conditions system (BICON).

What to look for

Grain that:

- has a blackened and sooty appearance
- gives off a fishy smell
- crushes in hand producing a greasy black powder.

Where to look

Crops that can be affected:

- wheat
- durum wheat
- triticale.

Importers

Infested grain or imported agricultural equipment that is contaminated with infested grain pose the greatest risk of Karnal bunt making it to Australia.

Growers and grain handlers

- Since only a few grains in each wheat head are infected, it is easiest to detect symptoms after the grain has been harvested.
- Check harvested grain regularly.
- Note any unpleasant or fishy smell.
- Look for discoloured grey or black seeds.

What to do

If you think you've found Karnal bunt in grain:

- do not disturb the infested produce (this may be as simple as closing the doors on a shipping container or sealing a silo)
- take a photo.
- collect a sample, if possible to do so without disturbing the infested produce.

Source: https://www.agriculture.gov.au/biosecurity-trade/pests-diseases-weeds/plant/identify/karnal-bunt